1. Create a table called Employee table.

SLNO	NAME	DESIGNATION	BRANCH
1	KRISHNA	ASSISTANT	CHENNAI
2	RAM	MANAGER	HYDERABAD
3	ASWIN	SUPERVISOR	KERALA
4	SUMAYYA	SUPERVISOR	MANGLOORE

a) Insert records into the table.

b) Add salary column to the table.

c) Alter the table by modifying column name.

d) Drop salary column of the Employee table.

e) Delete the rows of Employee table whose branch is “Chennai”.

2. Create Student table with columns Student_id, Student name with the following constraints on the Student table: • Student_id - Primary Key

3. Create table Department with columns Department_ID, Name, Student_ID with the following constraints on Department Table:
   • Department_ID - Primary Key
   • Student_ID - references Student table.

4. Create a table Employee with the following details

EMPNO	NAME	DESIGNATION	BRANCH
E211	SANJAY	MANAGER	CHENNAI
E364	RAM	SUPERVISOR	CHENNAI
E453	ASWIN	ASSISTANT	TRICHY
E896	SUMAYYA	ASSISTANT	MADURAI
E231	LAKSHMI	MANAGER	MADURAI

Perform the following:

a) Alter the table by adding a column salary.

b) Alter the table by modifying the column Name.

c) Describe the table Employee.

d) Truncate the table.

e) Delete the second row from the table.

f) Drop the table

5. Create a table Employees (Employee_Id, First_Name, Last_Name, Email, Phone_Number,Hire_Date, Job_Id, Salary, Commission_Pct, Manager_Id, Department_Id).

   a) Find out the employee_id, names, salaries of all the employees.

   b) List out the employees who work under Manager M215.

   c) Find the names of the employees who have a salary greater than or equal to 7500.

   d) List out the employees whose last name is 'MICHAEL'.

   e) Find the names of the employees who work in departments C21, F70, and H80.

   f) Display the unique Manager_Id.

6. Create a Sales table with the following fields (SalesId, Salesmenname, Branch, Salesamount, DOB).
   a) Insert five records.
   b) Calculate total salesamount in each branch.
   c) Calculate average salesamount in each branch.
   d) Display all the salesmen, DOB who are born in the month of December as day in
   character format i.e. 21-Dec-09.
   e) Display the name and DOB of salesmen in alphabetical order of the month.

7. Create Client _master with the following fields (ClientNO, Name, Address, City, State,Bal_due).

   a) Insert five records.

b) Find the names of clients whose bal_due > 5000.
c) Change the bal_due of ClientNo “C510” to Rs. 5100.
d) Change the name of Client_master to Clienttable.
e) Display the bal_due as “BALANCE”.

8. Create a Teacher table with the following fields (Name, DeptNo, Date_of_joining, DeptName, Location, Salary).
   a) Insert five records.
   b) Give increment of 25% salary for Mathematics Department.
   c) Perform Rollback command.
   d) Give increment of 15% salary for Commerce Department.
   e) Perform commit command.

9. Create an Emp table with the following fields (EmpNo, EmpName, Job, Basic, DA, HRA,PF, GrossPay, NetPay). Calculate DA as 30% of Basic and HRA as 40% of Basic.

   a) Insert five records and calculate GrossPay and NetPay.

   b) Display the employees whose Basic is lowest in each department.

   c) If NetPay is less than Rs. 10000, add Rs. 1200 as Special Allowances.

   d) Display the employees whose GrossPay lies between 10000 and 20000.

   e) Display all the employees who earn maximum salary.

10. Consider the Department table

Department	**Dname**	**Location**

a) Rename the table department as dept_table.
b) Add a new column PINCODE with not null constraints to the existing table Department.
c) Rename the column DNAME to DEPT_NAME in department table.
d) Change the data type of column location as CHAR with size 10.
e) Delete table.

11. Consider the Employee table.

EMP NO	NAME	DEPT	SALARY	DATOFJOIN	DESIG	BRANCH
E211	KRISHNA	MEDIA	25000	19-JAN-2024	ASSISTANT	CHENNAI
E364	RAM	IT	50000	25-NOV-2020	MANAGER	HYDERABAD
E453	ASWIN	CIVIL	65000	04-MAY-2010	SUPERVISOR	KERALA
E896	SUMAYYA	HR	62000	05-OCT-2018	SUPERVISOR	MANGLOORE
E125	ASWIN	FINANCE	180000	09-SEPT-2012	MANAGER	KERALA
E467	SUMAYYA	IT	150000	15-JUL-2005	MANAGER	MANGLOORE

Perform the following:
a) Display all the fields of the employee table.
b) Retrieve employee number and their salary.
c) Retrieve average salary of all employees.
d) Retrieve number of employees.
e) Retrieve distinct number of employees.
f) Retrieve total salary of employees group by employee name and count similar names.
g) Retrieve total salary of employee which is greater than 120000.
h) Display name of employee in descending order.
i) Display details of employees whose name is ASWIN and salary greater than 150000.

12. Create a table EMPLOYEE with the following schema:(Emp_no, E_name, E_address, E_ph_no, Email_id, Dept_no, Dept_name, Job_id,Designation, Salary, Date_of_join)
Write the SQL statements for the following queries.
a) List the E_no, E_name, Salary of all employees working for MANAGER.
b) Display all the details of the employees whose salary is more than the salary of any HR PROFESSIONAL.
c) List the employees in the ascending order of Designations of those joined after 2014.
d) List the employees along with their Experience and Salary.
e) List the employees who are either ‘ASSISTANT’ or ‘ANALYST’.

f) List the employees who joined on 1-MAY-2020, 3-DEC-1999, 17-DEC-2000, 19-JAN-80.
g) List the employees who are working for the Deptno D54 or D80.
h) List the Enames that start with 'A'.
i) Display the name as well as the five characters of name(s) starting with S.
j) List all the employees except 'PRESIDENT' & 'MANAGER' in ascending order of salaries.

13. Create the tables: Student (Sid, sname, sex, dob, dno) Department (dno, dname) Faculty (F_id, fname, designation, salary, dno)
a) Insert at least 5 tuples into each table.
b) List the details of students in the ascending order of date of birth.
c) Display the details of students from computer department.
d) List the faculties in the descending order of salary.
e) Display the total number of students in each department.
f) Display the total number of faculties in each department with salary greater than 25000.

14. Consider the following table.
Perform the following:
a) Write a SQL statement to display salesman_id, name, city, and commission who gets the commission within the range more than 0.10% and less than 0.12%.
b) Write a SQL statement to find those salesmen with all information who come from either Paris or Rome.
c) Write a SQL statement to find those salesmen with all information and name started with any letter within 'A' and 'K'.

15. Consider the following schema:
Sailors (sid, sname, rating, age)
Boats (bid, bname, color)
Reserves (sid, bid, day(date))
a) Find all the information of sailors who have reserved boat number 101.
b) Find the name of the boat reserved by Bob.
c) Find the names of sailors who have reserved a red boat, and list in order of age.
d) Find the names of sailors who have reserved at least one boat.
e) Find the ids and names of sailors who have reserved two different boats on the same day.
f) Find the ids of sailors who have reserved a red boat or a green boat.
g) Find the name and the age of the youngest sailor.
h) Count the number of different sailor names.
i) Find the average age of sailors for each rating level.
j) Find the average age of sailors for each rating level that has at least two sailors.

PL/SQL

16. Write a pl/sql function to Check whether number is prime or not.
17. Write a pl/sql function to Find the Factorial of a number.
18. Write a pl/sql Function to get the sum of 2 numbers
19. Write a pl/sql function to return the rating of a sailor on giving an sid.
20. Write a pl/sql function to print Weekday in letters.
21. Write a PL/SQL program to perform banking operations using procedures. create table account(account_no number as primary key,account_holder Varchar2(20), balance number)
22. Write a pl/sql function to check whether the given string is a palindrome or not.
23. Write a pl/sql function that computes and returns the maximum of two values.
24. Write a pl/sql procedure to find the sum of two numbers
25. Write a pl/sql procedure to insert the roll no. and name of a student into the STUDENT table.
26. Write a pl/sql procedure to get the count of instructors in the given department
27. Create a cursor to Display sid and rating of all sailors
28. Create a cursor to display the details of employees.
29. Write a trigger to restrict deletion operation on Professor table.
30. Create a trigger to manage the counts of male and female students.
31. Write the MongoDB commands to create a collection named Employee. Create a collection named empDetails You can use any of the fields Empid,Name, Dept, Age , e_mail, phone,salary,city.
32. Write the MongoDB commands to insert the details of the multiple Employees (at least 5) in the form of documents in the employee collection.
33. Write the MongoDB commands to retrieve the fields empid, name, dept,phone, salary, city for all the documents in the collection employee
34. Write the MongoDB commands to find the details of employee whose name is ’ mohan’
35. Write the MongoDB commands to fetch the documents of employees whose salary >=5000
36. Write the MongoDB commands to find the documents of employees whose name starts with letter r
37. Write the MongoDB commands to retrieve the number of employees per department from the employee collection
38. Write the MongoDB commands to find the documents of employees whose name is not in ‘mohan’ , ‘raju’, ‘bhuvan’
39. Write the MongoDB commands to retrieve the details of employees whose age is less than 30. Display only the fields name, salary.
40. Write the MongoDB commands to find the details of employees whose salary is >5000 and age is < 30

41. Write the MongoDB commands to arrange the name of the employees in ascending order along with all the columns
42. Write the MongoDB commands to delete all the documents of employees whose age>56